

Task 1:

```
root@LAPTOP-0PA61QEN:~# ls /home
alurusaranya
root@LAPTOP-0PA61QEN:~# su alurusaranya
alurusaranya@LAPTOP-0PA61QEN:/root$ ls
ls: cannot open directory '.': Permission denied
alurusaranya@LAPTOP-0PA61QEN:/root$ cd ~
alurusaranya@LAPTOP-0PA61QEN:~$ pwd
/home/alurusaranya
alurusaranya@LAPTOP-0PA61QEN:~$ nano fork_exec.c
alurusaranya@LAPTOP-0PA61QEN:~$ gcc fork_exec.c -o fork_exec
alurusaranya@LAPTOP-0PA61QEN:~$ ./fork_exec
Enter a Linux command: ls

Parent Process
Parent PID : 462
Child PID  : 463

Child Process
Child PID : 463
Parent PID: 462

Executing command: ls
OSSP fork_exec fork_exec.c hello hello.c

Child process completed.
alurusaranya@LAPTOP-0PA61QEN:~$ |
```

```
GNU nano 7.2                                fork_exec.c *
#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>
#include <sys/types.h>
#include <sys/wait.h>

int main() {
    char command[100];
    pid_t pid;

    printf("Enter a Linux command: ");
    scanf("%s", command);

    pid = fork();

    if (pid < 0) {
        printf("Fork failed!\n");
        return 1;
    }

    if (pid == 0) {
        // Child process
        printf("\nChild Process\n");
        printf("Child PID : %d\n", getpid());
        printf("Parent PID: %d\n", getppid());

        printf("\nExecuting command: %s\n", command);

        execlp(command, command, NULL);

        // Executes only if execlp() fails
        perror("exec failed");
        exit(1);
    }
    else {
        // Parent process
        printf("\nParent Process\n");
        printf("Parent PID : %d\n", getpid());
        printf("Child PID  : %d\n", pid);

        wait(NULL);

        printf("\nChild process completed.\n");
    }

    return 0;
}
```

[illegible]

## Virtualization features:

Virtualization: VT-x  
 Hypervisor vendor: Microsoft  
 Virtualization type: full

## Caches (sum of all):

L1d: 288 KiB (6 instances)  
 L1i: 384 KiB (6 instances)  
 L2: 18 MiB (6 instances)  
 L3: 12 MiB (1 instance)

## NUMA:

NUMA node(s): 1  
 NUMA node0 CPU(s): 0-5

## Vulnerabilities:

Gather data sampling: Not affected  
 Ghostwrite: Not affected  
 Indirect target selection: Not affected  
 Itlb multihit: Not affected  
 L1tf: Not affected  
 Mds: Not affected  
 Meltdown: Not affected  
 Mmio stale data: Not affected  
 Old microcode: Not affected  
 Reg file data sampling: Not affected  
 Retbleed: Mitigation; Enhanced IBRS  
 Spec rstack overflow: Not affected  
 Spec store bypass: Mitigation; Speculative Store Bypass disabled via prctl  
 Spectre v1: Mitigation; usercopy/swapgs barriers and \_\_user pointer sanitization  
 Spectre v2: Mitigation; Enhanced / Automatic IBRS; IBPB conditional; PBR SB-eIBRS; STIBP; BHI BHI\_DIS\_S  
 Srbds: Not affected  
 Tsa: Not affected  
 Tsx async abort: Not affected  
 Vmscape: Not affected  
 vmscape: not affected

alurusaranya@LAPTOP-0PA61QEN:~\$ ps

```
PID TTY      TIME CMD
540 pts/0    00:00:00 bash
560 pts/0    00:00:00 ps
```

```

380 pts/0    00:00:00 ps
alurusaranya@LAPTOP-0PA61QEN:~$ top
top - 05:59:52 up 6 min,  2 users,  load average: 0.00, 0.04, 0
Tasks:  26 total,   1 running,  25 sleeping,   0 stopped,   0 z
%Cpu(s):  0.0 us,  0.0 sy,  0.0 ni,100.0 id,  0.0 wa,  0.0 hi,
MiB Mem :   7712.1 total,   6502.2 free,    491.1 used,    868.
MiB Swap:   2048.0 total,   2048.0 free,     0.0 used.   7221.

```

| PID | USER     | PR | NI | VIRT    | RES   | SHR   | S | %CPU | %MEM |
|-----|----------|----|----|---------|-------|-------|---|------|------|
| 1   | root     | 20 | 0  | 21652   | 13072 | 9732  | S | 0.0  | 0.2  |
| 2   | root     | 20 | 0  | 3180    | 2200  | 2072  | S | 0.0  | 0.0  |
| 6   | root     | 20 | 0  | 3196    | 2128  | 2008  | S | 0.0  | 0.0  |
| 52  | root     | 19 | -1 | 50360   | 16360 | 15284 | S | 0.0  | 0.2  |
| 105 | root     | 20 | 0  | 25604   | 6964  | 5128  | S | 0.0  | 0.1  |
| 111 | systemd+ | 20 | 0  | 21460   | 13272 | 11016 | S | 0.0  | 0.2  |
| 112 | systemd+ | 20 | 0  | 91028   | 7968  | 6996  | S | 0.0  | 0.1  |
| 164 | root     | 20 | 0  | 4244    | 2684  | 2432  | S | 0.0  | 0.0  |
| 165 | message+ | 20 | 0  | 9592    | 5328  | 4624  | S | 0.0  | 0.1  |
| 172 | root     | 20 | 0  | 18148   | 8920  | 7848  | S | 0.0  | 0.1  |
| 175 | root     | 20 | 0  | 1756104 | 13076 | 10888 | S | 0.0  | 0.2  |
| 193 | syslog   | 20 | 0  | 222516  | 5968  | 4556  | S | 0.0  | 0.1  |
| 199 | root     | 20 | 0  | 3124    | 1976  | 1836  | S | 0.0  | 0.0  |
| 232 | root     | 20 | 0  | 107016  | 23412 | 13948 | S | 0.0  | 0.3  |
| 288 | root     | 20 | 0  | 6672    | 4496  | 3728  | S | 0.0  | 0.1  |
| 333 | root     | 20 | 0  | 20312   | 11576 | 9452  | S | 0.0  | 0.1  |
| 334 | root     | 20 | 0  | 21164   | 3512  | 1796  | S | 0.0  | 0.0  |
| 355 | root     | 20 | 0  | 6080    | 5324  | 3696  | S | 0.0  | 0.1  |
| 476 | root     | 20 | 0  | 3188    | 1112  | 980   | S | 0.0  | 0.0  |
| 477 | root     | 20 | 0  | 3204    | 1252  | 1108  | S | 0.0  | 0.0  |
| 481 | root     | 20 | 0  | 6080    | 5336  | 3664  | S | 0.0  | 0.1  |
| 514 | root     | 20 | 0  | 6808    | 4444  | 3980  | S | 0.0  | 0.1  |
| 517 | alurusa+ | 20 | 0  | 20316   | 11544 | 9412  | S | 0.0  | 0.1  |
| 518 | alurusa+ | 20 | 0  | 21160   | 3580  | 1852  | S | 0.0  | 0.0  |
| 540 | alurusa+ | 20 | 0  | 6084    | 5300  | 3664  | S | 0.0  | 0.1  |
| 561 | alurusa+ | 20 | 0  | 9292    | 5412  | 3240  | R | 0.0  | 0.1  |